

Lecture 2: Number Fields & Kronecker–Weber

Mini-Course: Magma and the LMFDB

Eran Assaf

CTNT Summer School 2026

MIT

Lecture Overview

Number Fields and Their Invariants

Abelian Extensions and Kronecker–Weber

Class Groups, Units, and Cohen–Lenstra

Hilbert Class Field and the LMFDB

Number Fields and Their Invariants

What is a Number Field?

A **number field** K is a finite extension of $\mathbb{Q}$.

Examples:

- Quadratic: $\mathbb{Q}(\sqrt{d})$ for $d \in \mathbb{Z}$ squarefree.
- Cyclotomic: $\mathbb{Q}(\zeta_n)$, $\zeta_n = e^{2\pi i/n}$.
- Composite: $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, ...

What is a Number Field?

A **number field** K is a finite extension of $\mathbb{Q}$.

Examples:

- Quadratic: $\mathbb{Q}(\sqrt{d})$ for $d \in \mathbb{Z}$ squarefree.
- Cyclotomic: $\mathbb{Q}(\zeta_n)$, $\zeta_n = e^{2\pi i/n}$.
- Composite: $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, ...

First invariants:

- Ring of **integers** $\mathbb{Z}_K = \{\alpha \in K : \alpha \text{ integral over } \mathbb{Z}\}$.
- **Signature** (r_1, r_2) : r_1 real and r_2 pairs of complex embeddings.
- **Discriminant** $\Delta_K = \text{disc}(\mathbb{Z}_K) \in \mathbb{Z}$ (next slide).

The Discriminant via the Trace Form

Lecture 1 callback. $\mathbb{Z}_K \subset K$ is an integral $\mathbb{Z}$ -lattice in the sense of Lecture 1, equipped with the bilinear form $(\alpha, \beta) \mapsto \text{Tr}_{K/\mathbb{Q}}(\alpha\beta) : K \times K \rightarrow \mathbb{Q}$. Then $\Delta_K = \text{disc}(\mathbb{Z}_K)$ – the lattice discriminant.

The Discriminant via the Trace Form

Lecture 1 callback. $\mathbb{Z}_K \subset K$ is an integral $\mathbb{Z}$ -lattice in the sense of Lecture 1, equipped with the bilinear form $(\alpha, \beta) \mapsto \text{Tr}_{K/\mathbb{Q}}(\alpha\beta) : K \times K \rightarrow \mathbb{Q}$. Then $\Delta_K = \text{disc}(\mathbb{Z}_K)$ – the lattice discriminant.

Example (Quadratic fields)

For $F = \mathbb{Q}(\sqrt{d})$, $d \neq 1$ squarefree, $\mathbb{Z}_F = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$ if $d \equiv 2, 3 \pmod{4}$, and $\mathbb{Z} \oplus \mathbb{Z} \frac{1+\sqrt{d}}{2}$ if $d \equiv 1 \pmod{4}$.

The Discriminant via the Trace Form

Lecture 1 callback. $\mathbb{Z}_K \subset K$ is an integral $\mathbb{Z}$ -lattice in the sense of Lecture 1, equipped with the bilinear form $(\alpha, \beta) \mapsto \text{Tr}_{K/\mathbb{Q}}(\alpha\beta) : K \times K \rightarrow \mathbb{Q}$. Then $\Delta_K = \text{disc}(\mathbb{Z}_K)$ – the lattice discriminant.

Example (Quadratic fields)

For $F = \mathbb{Q}(\sqrt{d})$, $d \neq 1$ squarefree, $\mathbb{Z}_F = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$ if $d \equiv 2, 3 \pmod{4}$, and $\mathbb{Z} \oplus \mathbb{Z} \frac{1+\sqrt{d}}{2}$ if $d \equiv 1 \pmod{4}$.

The Gram matrix of $\text{Tr}(\alpha\beta)$ is

$$\begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 2 & 1 \\ 1 & \frac{1+d}{2} \end{pmatrix},$$

giving $\Delta_F = 4d$ or d , respectively.

The Discriminant via the Trace Form

Lecture 1 callback. $\mathbb{Z}_K \subset K$ is an integral $\mathbb{Z}$ -lattice in the sense of Lecture 1, equipped with the bilinear form $(\alpha, \beta) \mapsto \text{Tr}_{K/\mathbb{Q}}(\alpha\beta) : K \times K \rightarrow \mathbb{Q}$. Then $\Delta_K = \text{disc}(\mathbb{Z}_K)$ – the lattice discriminant.

Example (Quadratic fields)

For $F = \mathbb{Q}(\sqrt{d})$, $d \neq 1$ squarefree, $\mathbb{Z}_F = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}$ if $d \equiv 2, 3 \pmod{4}$, and $\mathbb{Z} \oplus \mathbb{Z} \frac{1+\sqrt{d}}{2}$ if $d \equiv 1 \pmod{4}$.

The Gram matrix of $\text{Tr}(\alpha\beta)$ is

$$\begin{pmatrix} 2 & 0 \\ 0 & 2d \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 2 & 1 \\ 1 & \frac{1+d}{2} \end{pmatrix},$$

giving $\Delta_F = 4d$ or d , respectively.

Theorem (Ramification)

A prime p ramifies in $\mathbb{Z}_K$ iff $p \mid \Delta_K$.

Running Example: A Cubic of Discriminant -23

Let $K = \mathbb{Q}(\alpha)$ where $\alpha^3 - \alpha - 1 = 0$.

```
1 > _<x> := PolynomialRing(Rationals());  
2 > K<a> := NumberField(x^3 - x - 1);  
3 > ZK := Integers(K);  
4 > Discriminant(ZK);  
5 -23  
6 > Signature(K);  
7 1 1  
8 > ClassNumber(K);  
9 1
```

Running Example: A Cubic of Discriminant -23

Let $K = \mathbb{Q}(\alpha)$ where $\alpha^3 - \alpha - 1 = 0$.

```
1 > _<x> := PolynomialRing(Rationals());  
2 > K<a> := NumberField(x^3 - x - 1);  
3 > ZK := Integers(K);  
4 > Discriminant(ZK);  
5 -23  
6 > Signature(K);  
7 1 1  
8 > ClassNumber(K);  
9 1
```

Why this field?

- Smallest $|\Delta_K|$ among cubic fields.
- Galois closure has group S_3 – *not abelian*, so K-W does not apply.
- LMFDB label: **3.1.23.1**.
- *Watch this field – it will reappear in a surprising place at the end of the lecture.*

Splitting of Primes in $\mathbb{Q}[x]/(x^3 - x - 1)$

Patterns in $\mathbb{Z}_K$ of the cubic field $K = \mathbb{Q}[x]/(x^3 - x - 1)$:

```
1 > for p in [2, 3, 5, 23, 59] do
2 >   F := Factorization(p*ZK);
3 >   printf "p = %o: %o\n", p, [<Norm(t[1]), t[2]> : t in F];
4 > end for;
5 p = 2: [ <8, 1> ]           // inert
6 p = 3: [ <27, 1> ]          // inert
7 p = 5: [ <5, 1>, <25, 1> ]   // partially split
8 p = 23: [ <23, 2>, <23, 1> ] // ramified (p | disc)
9 p = 59: [ <59, 1>, <59, 1>, <59, 1> ] // totally split
```

Splitting of Primes in $\mathbb{Q}[x]/(x^3 - x - 1)$

Patterns in $\mathbb{Z}_K$ of the cubic field $K = \mathbb{Q}[x]/(x^3 - x - 1)$:

```
1 > for p in [2, 3, 5, 23, 59] do
2 >   F := Factorization(p*ZK);
3 >   printf "p = %o: %o\n", p, [<Norm(t[1]), t[2]> : t in F];
4 > end for;
5 p = 2: [ <8, 1> ]           // inert
6 p = 3: [ <27, 1> ]          // inert
7 p = 5: [ <5, 1>, <25, 1> ]   // partially split
8 p = 23: [ <23, 2>, <23, 1> ] // ramified (p | disc)
9 p = 59: [ <59, 1>, <59, 1>, <59, 1> ] // totally split
```

Theory. For $p \nmid \Delta_K$, the splitting of p in K matches the factorization of $x^3 - x - 1 \bmod p$ (Kummer–Dedekind).

Abelian Extensions and Kronecker–Weber

In our cubic, the splitting of primes is governed by $x^3 - x - 1 \bmod p$ – a polynomial condition.

In our cubic, the splitting of primes is governed by $x^3 - x - 1 \bmod p$ – a polynomial condition.

When $K/\mathbb{Q}$ is abelian, something stronger happens: the splitting of p depends only on p modulo some fixed integer.

From General to Abelian

In our cubic, the splitting of primes is governed by $x^3 - x - 1 \bmod p$ – a polynomial condition.

When $K/\mathbb{Q}$ is abelian, something stronger happens: the splitting of p depends only on p modulo some fixed integer.

This dramatic simplification is the content of *Kronecker–Weber*, and it leads to a complete classification of abelian extensions of $\mathbb{Q}$.

The Kronecker–Weber Theorem

Theorem (Kronecker, Weber)

Every finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_n)$.

The Kronecker–Weber Theorem

Theorem (Kronecker, Weber)

Every finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_n)$.

Strategy. To understand abelian extensions of $\mathbb{Q}$, study the cyclotomic tower.

The Kronecker–Weber Theorem

Theorem (Kronecker, Weber)

Every finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_n)$.

Strategy. To understand abelian extensions of $\mathbb{Q}$, study the cyclotomic tower.

Examples:

$$\mathbb{Q}(i) = \mathbb{Q}(\zeta_4),$$

$$\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\zeta_8), \quad \text{since } \sqrt{2} = \zeta_8 + \zeta_8^{-1},$$

$$\mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\zeta_5), \quad \text{since } \sqrt{5} = 2(\zeta_5 + \zeta_5^{-1}) + 1.$$

The Kronecker–Weber Theorem

Theorem (Kronecker, Weber)

Every finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_n)$.

Strategy. To understand abelian extensions of $\mathbb{Q}$, study the cyclotomic tower.

Examples:

$$\mathbb{Q}(i) = \mathbb{Q}(\zeta_4),$$

$$\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\zeta_8), \quad \text{since } \sqrt{2} = \zeta_8 + \zeta_8^{-1},$$

$$\mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\zeta_5), \quad \text{since } \sqrt{5} = 2(\zeta_5 + \zeta_5^{-1}) + 1.$$

Question: for a given abelian K , what is the *smallest* n ?

Frobenius and the Conductor

Suppose $K/\mathbb{Q}$ is abelian. For each unramified prime p , there is a **Frobenius element** $\text{Frob}_p \in \text{Gal}(K/\mathbb{Q})$.

(In general, Frob_p is only a conjugacy class. The abelian hypothesis makes it well-defined as an element.)

Frobenius and the Conductor

Suppose $K/\mathbb{Q}$ is abelian. For each unramified prime p , there is a **Frobenius element** $\text{Frob}_p \in \text{Gal}(K/\mathbb{Q})$.

(In general, Frob_p is only a conjugacy class. The abelian hypothesis makes it well-defined as an element.)

Theorem (Reciprocity, weak form)

*If $K/\mathbb{Q}$ is abelian, there exists a smallest integer $f(K) \in \mathbb{Z}_{\geq 1}$ — the **conductor** — such that Frob_p depends only on $p \bmod f(K)$ (for $p \nmid f(K)$).*

Frobenius and the Conductor

Suppose $K/\mathbb{Q}$ is abelian. For each unramified prime p , there is a **Frobenius element** $\text{Frob}_p \in \text{Gal}(K/\mathbb{Q})$.

(In general, Frob_p is only a conjugacy class. The abelian hypothesis makes it well-defined as an element.)

Theorem (Reciprocity, weak form)

*If $K/\mathbb{Q}$ is abelian, there exists a smallest integer $f(K) \in \mathbb{Z}_{\geq 1}$ — the **conductor** — such that Frob_p depends only on $p \bmod f(K)$ (for $p \nmid f(K)$).*

Theorem (Kronecker–Weber, refined)

$K \subseteq \mathbb{Q}(\zeta_n)$ if and only if $f(K) \mid n$. In particular, $K \subseteq \mathbb{Q}(\zeta_{f(K)})$, the smallest cyclotomic field containing K .

Frobenius and the Conductor

Suppose $K/\mathbb{Q}$ is abelian. For each unramified prime p , there is a **Frobenius element** $\text{Frob}_p \in \text{Gal}(K/\mathbb{Q})$.

(In general, Frob_p is only a conjugacy class. The abelian hypothesis makes it well-defined as an element.)

Theorem (Reciprocity, weak form)

*If $K/\mathbb{Q}$ is abelian, there exists a smallest integer $f(K) \in \mathbb{Z}_{\geq 1}$ — the **conductor** — such that Frob_p depends only on $p \bmod f(K)$ (for $p \nmid f(K)$).*

Theorem (Kronecker–Weber, refined)

$K \subseteq \mathbb{Q}(\zeta_n)$ if and only if $f(K) \mid n$. In particular, $K \subseteq \mathbb{Q}(\zeta_{f(K)})$, the smallest cyclotomic field containing K .

The agreement of these two characterizations of $f(K)$ is itself part of the content of Kronecker–Weber.

Conductor: Examples

Example (Cyclotomic fields)

For $F = \mathbb{Q}(\zeta_n)$ with $n \not\equiv 2 \pmod{4}$, $\text{Gal}(F/\mathbb{Q}) \simeq (\mathbb{Z}/n\mathbb{Z})^\times$, $\sigma_a(\zeta_n) = \zeta_n^a$, and $\text{Frob}_p = \sigma_p$. The conductor is $f(F) = n$.

Conductor: Examples

Example (Cyclotomic fields)

For $F = \mathbb{Q}(\zeta_n)$ with $n \not\equiv 2 \pmod{4}$, $\text{Gal}(F/\mathbb{Q}) \simeq (\mathbb{Z}/n\mathbb{Z})^\times$, $\sigma_a(\zeta_n) = \zeta_n^a$, and $\text{Frob}_p = \sigma_p$. The conductor is $\mathfrak{f}(F) = n$.

Example (Quadratic fields)

For $F = \mathbb{Q}(\sqrt{d})$ squarefree with $\Delta_F \in \{d, 4d\}$, $\text{Gal}(F/\mathbb{Q}) \simeq \{\pm 1\}$, and $\text{Frob}_p = \left(\frac{\Delta_F}{p}\right)$. The conductor is $\mathfrak{f}(F) = |\Delta_F|$ – exactly the discriminant we computed before.

Conductor: Examples

Example (Cyclotomic fields)

For $F = \mathbb{Q}(\zeta_n)$ with $n \not\equiv 2 \pmod{4}$, $\text{Gal}(F/\mathbb{Q}) \simeq (\mathbb{Z}/n\mathbb{Z})^\times$, $\sigma_a(\zeta_n) = \zeta_n^a$, and $\text{Frob}_p = \sigma_p$. The conductor is $\mathfrak{f}(F) = n$.

Example (Quadratic fields)

For $F = \mathbb{Q}(\sqrt{d})$ squarefree with $\Delta_F \in \{d, 4d\}$, $\text{Gal}(F/\mathbb{Q}) \simeq \{\pm 1\}$, and $\text{Frob}_p = \left(\frac{\Delta_F}{p}\right)$. The conductor is $\mathfrak{f}(F) = |\Delta_F|$ – exactly the discriminant we computed before.

Slogan. The conductor is the smallest modulus on which the splitting behavior of primes in K depends only on residue classes.

Explicit Kronecker–Weber

Start with K abstract: $K = \mathbb{Q}[x]/(x^3 + x^2 - 2x - 1)$.

```
1 > _<x> := PolynomialRing(Rationals());
2 > K<a> := NumberField(x^3 + x^2 - 2*x - 1);
3 > Discriminant(Integers(K));
4 49
5 > IsAbelian(K);
6 true
7 > // Conductor: smallest n with K c Q(zeta_n)
8 > Minimum(Conductor(AbelianExtension(K)));
9 7
```

Explicit Kronecker–Weber

Start with K abstract: $K = \mathbb{Q}[x]/(x^3 + x^2 - 2x - 1)$.

```
1 > _<x> := PolynomialRing(Rationals());  
2 > K<a> := NumberField(x^3 + x^2 - 2*x - 1);  
3 > Discriminant(Integers(K));  
4 49  
5 > IsAbelian(K);  
6 true  
7 > // Conductor: smallest n with K c Q(zeta_n)  
8 > Minimum(Conductor(AbelianExtension(K)));  
9 7
```

K-W in action. The conductor predicts $K \subset \mathbb{Q}(\zeta_7)$ – let us realize the embedding explicitly.

```
1 > Q7<z7> := CyclotomicField(7);  
2 > bool, embed := IsSubfield(K, Q7); bool;  
3 true  
4 > assert embed(a) eq z7 + z7^(-1);
```

Explicit Kronecker–Weber

Start with K abstract: $K = \mathbb{Q}[x]/(x^3 + x^2 - 2x - 1)$.

```
1 > _<x> := PolynomialRing(Rationals());  
2 > K<a> := NumberField(x^3 + x^2 - 2*x - 1);  
3 > Discriminant(Integers(K));  
4 49  
5 > IsAbelian(K);  
6 true  
7 > // Conductor: smallest n with K c Q(zeta_n)  
8 > Minimum(Conductor(AbelianExtension(K)));  
9 7
```

K-W in action. The conductor predicts $K \subset \mathbb{Q}(\zeta_7)$ – let us realize the embedding explicitly.

```
1 > Q7<z7> := CyclotomicField(7);  
2 > bool, embed := IsSubfield(K, Q7); bool;  
3 true  
4 > assert embed(a) eq z7 + z7^(-1);
```

Gauss period. $a = \zeta_7 + \zeta_7^{-1}$ – the sum over the index-3 subgroup of $(\mathbb{Z}/7\mathbb{Z})^\times$.

Galois Groups: Beyond the Abelian Case

Our running cubic $K = \mathbb{Q}[x]/(x^3 - x - 1)$ is *not* abelian over $\mathbb{Q}$.

```
1 > K<a> := NumberField(x^3 - x - 1);
2 > IsAbelian(K);
3 false
4 > L := NormalClosure(K);
5 > G := GaloisGroup(L);
6 > #G;
7 6
8 > assert IsIsomorphic(G, Sym(3));
9 > // L = K(sqrt(disc K)) = K(sqrt(-23))
10 > Discriminant(Integers(K));
11 -23
```


Galois Groups: Beyond the Abelian Case

Our running cubic $K = \mathbb{Q}[x]/(x^3 - x - 1)$ is *not* abelian over $\mathbb{Q}$.

```
1 > K<a> := NumberField(x^3 - x - 1);
2 > IsAbelian(K);
3 false
4 > L := NormalClosure(K);
5 > G := GaloisGroup(L);
6 > #G;
7 6
8 > assert IsIsomorphic(G, Sym(3));
9 > // L = K(sqrt(disc K)) = K(sqrt(-23))
10 > Discriminant(Integers(K));
11 -23
```

Observation. $\tilde{K} = K(\sqrt{-23})$, $\text{Gal}(\tilde{K}/\mathbb{Q}) \simeq S_3$. The unique abelian subextension is $\mathbb{Q}(\sqrt{-23})$, with conductor $f(\mathbb{Q}(\sqrt{-23})) = 23$.

Galois Groups: Beyond the Abelian Case

Our running cubic $K = \mathbb{Q}[x]/(x^3 - x - 1)$ is *not* abelian over $\mathbb{Q}$.

```
1 > K<a> := NumberField(x^3 - x - 1);
2 > IsAbelian(K);
3 false
4 > L := NormalClosure(K);
5 > G := GaloisGroup(L);
6 > #G;
7 6
8 > assert IsIsomorphic(G, Sym(3));
9 > // L = K(sqrt(disc K)) = K(sqrt(-23))
10 > Discriminant(Integers(K));
11 -23
```

Observation. $\tilde{K} = K(\sqrt{-23})$, $\text{Gal}(\tilde{K}/\mathbb{Q}) \simeq S_3$. The unique abelian subextension is $\mathbb{Q}(\sqrt{-23})$, with conductor $f(\mathbb{Q}(\sqrt{-23})) = 23$.

Bridge to non-abelian CFT. K-W generalizes to imaginary quadratic base fields via CM theory; this is Kronecker's Jugendtraum, which we will talk about in Lecture 3.

Class Groups, Units, and Cohen–Lenstra

Class Groups and Units

The **class group** $\text{Cl}(K) = I_K/P_K = I_K/\mathfrak{I}(K^\times)$ measures the failure of unique factorization in $\mathbb{Z}_K$. Its order $h_K = \# \text{Cl}(K)$ is the **class number**.

Class Groups and Units

The **class group** $\text{Cl}(K) = I_K/P_K = I_K/\mathfrak{p}(K^\times)$ measures the failure of unique factorization in $\mathbb{Z}_K$. Its order $h_K = \# \text{Cl}(K)$ is the **class number**.

Theorem (Dirichlet's Unit Theorem)

If K has signature (r_1, r_2) , then $\mathbb{Z}_K^\times \simeq \mu(K) \times \mathbb{Z}^{r_1+r_2-1}$, where $\mu(K)$ is the (finite) group of roots of unity in K .

Class Groups and Units

The **class group** $\text{Cl}(K) = I_K/P_K = I_K/\mathfrak{I}(K^\times)$ measures the failure of unique factorization in $\mathbb{Z}_K$. Its order $h_K = \# \text{Cl}(K)$ is the **class number**.

Theorem (Dirichlet's Unit Theorem)

If K has signature (r_1, r_2) , then $\mathbb{Z}_K^\times \simeq \mu(K) \times \mathbb{Z}^{r_1+r_2-1}$, where $\mu(K)$ is the (finite) group of roots of unity in K .

Examples.

- $\mathbb{Q}(\sqrt{-1})$: $h = 1$ (PID), $\mathbb{Z}_K^\times = \mu_4$.
- $\mathbb{Q}(\sqrt{-5})$: $h = 2$, $\text{Cl}(K) \simeq \mathbb{Z}/2$ – UFD fails. (Lecture 1 callback: those two reduced binary forms of discriminant -20 , $x^2 + 5y^2$ and $2x^2 - 2xy + 3y^2$, were the two classes of $\text{Cl}(\mathbb{Q}(\sqrt{-5}))$.)
- $\mathbb{Q}(\sqrt{-21})$: $h = 4$, $\text{Cl}(K) \simeq \mathbb{Z}/2 \times \mathbb{Z}/2$ – structure carries information beyond h .
- $\mathbb{Q}(\sqrt{2})$: $h = 1$, but $\mathbb{Z}_K^\times = \langle -1 \rangle \times \langle 1 + \sqrt{2} \rangle$ – infinite unit group (real signature, $r_1 + r_2 - 1 = 1$).

Computing Class Groups and Units in Magma

```
1 > // Class group of  $\mathbb{Q}(\sqrt{-23})$ 
2 > K<b> := QuadraticField(-23);
3 > C, phi := ClassGroup(K);
4 > #C;
5 3
6 > // A non-principal ideal generating  $\text{Cl}(K)$ 
7 > [K!g : g in Generators(phi(C.1))];
8 [ 2, 1/2*(b + 1) ]
9 > // Units of  $\mathbb{Q}(\sqrt{2})$ 
10 > L<sqrt2> := QuadraticField(2);
11 > U, psi := UnitGroup(L);
12 > [L!psi(u) : u in Generators(U)];
13 [ -sqrt2 + 1, -1 ]
```

Computing Class Groups and Units in Magma

```
1 > // Class group of Q(sqrt(-23))
2 > K<b> := QuadraticField(-23);
3 > C, phi := ClassGroup(K);
4 > #C;
5 3
6 > // A non-principal ideal generating Cl(K)
7 > [K!g : g in Generators(phi(C.1))];
8 [ 2, 1/2*(b + 1) ]
9 > // Units of Q(sqrt(2))
10 > L<sqrt2> := QuadraticField(2);
11 > U, psi := UnitGroup(L);
12 > [L!psi(u) : u in Generators(U)];
13 [ -sqrt2 + 1, -1 ]
```

Fundamental unit. The infinite-order generator $1 + \sqrt{2}$ of $\mathbb{Z}_{\mathbb{Q}(\sqrt{2})}^\times$.

Algorithmic note. Class group computations are essentially instantaneous for small imaginary quadratic fields, but become hard (subexponential time) for general number fields of large discriminant.

Class Number Heuristics: Cohen–Lenstra

How are class groups distributed as K varies? Very little is known about the behavior of class numbers.

Class Number Heuristics: Cohen–Lenstra

How are class groups distributed as K varies? Very little is known about the behavior of class numbers.

Conjecture (Gauss, 1801)

There are infinitely many real quadratic fields K with $h_K = 1$.

Class Number Heuristics: Cohen–Lenstra

How are class groups distributed as K varies? Very little is known about the behavior of class numbers.

Conjecture (Gauss, 1801)

There are infinitely many real quadratic fields K with $h_K = 1$.

This is open, but Cohen and Lenstra (1984) made a much stronger prediction: $\text{Cl}(K)$ should behave like a random finite abelian group A , weighted by $1/\#\text{Aut}(A)$.

Class Number Heuristics: Cohen–Lenstra

How are class groups distributed as K varies? Very little is known about the behavior of class numbers.

Conjecture (Gauss, 1801)

There are infinitely many real quadratic fields K with $h_K = 1$.

This is open, but Cohen and Lenstra (1984) made a much stronger prediction: $\text{Cl}(K)$ should behave like a random finite abelian group A , weighted by $1/\#\text{Aut}(A)$.

Conjecture (Cohen–Lenstra, $h = 1$)

Among real quadratic fields of the form $K = \mathbb{Q}(\sqrt{p})$, with $p \equiv 1 \pmod{4}$, the proportion with $h_K = 1$ is

$$\prod_{k \geq 2} \frac{1}{\zeta(k)(1 - 2^{-k})} \approx 0.7545.$$

Class Number Heuristics: Cohen–Lenstra

How are class groups distributed as K varies? Very little is known about the behavior of class numbers.

Conjecture (Gauss, 1801)

There are infinitely many real quadratic fields K with $h_K = 1$.

This is open, but Cohen and Lenstra (1984) made a much stronger prediction: $\text{Cl}(K)$ should behave like a random finite abelian group A , weighted by $1/\#\text{Aut}(A)$.

Conjecture (Cohen–Lenstra, $h = 1$)

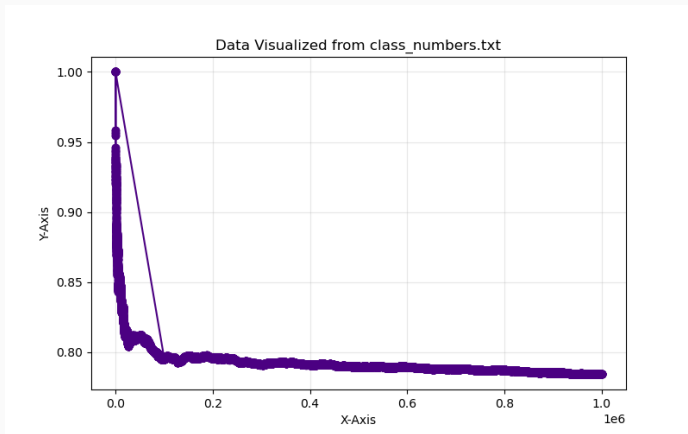
Among real quadratic fields of the form $K = \mathbb{Q}(\sqrt{p})$, with $p \equiv 1 \pmod{4}$, the proportion with $h_K = 1$ is

$$\prod_{k \geq 2} \frac{1}{\zeta(k)(1 - 2^{-k})} \approx 0.7545.$$

Question. Can we test this with Magma?

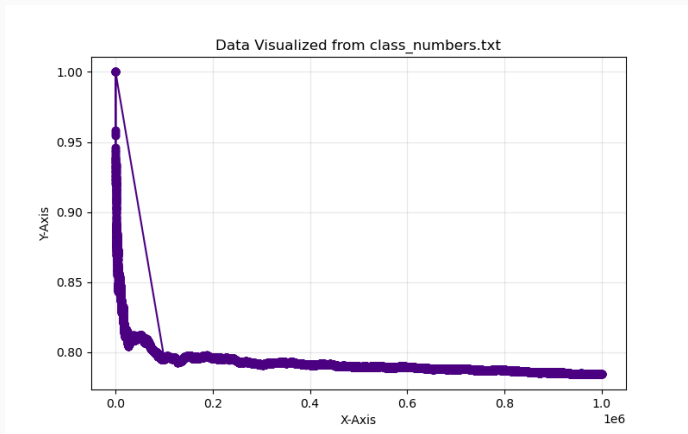
```
1 > B := 10^6;  
2 > primes := [p : p in PrimesUpTo(B) | p mod 4 eq 1];  
3 > // The next line takes 3-4 minutes -- pre-compute, then load!  
4 > class_nos := [ClassNumber(p) : p in primes];  
5 > // Multiplicity(L, v) counts occurrences of v in L  
6 > Multiplicity(class_nos,1) / #class_nos * 1.0;  
7 0.783969368219527760051052967454  
8 > ratio := func<n | Multiplicity(class_nos[1..n],1) / n * 1.0>;  
9 > ratios := [ratio(n) : n in [1..#class_nos]];  
10 > pts := [<primes[n], ratios[n]> : n in [1..#primes]];  
11 > PrintFile("class_numbers.txt", pts);
```

Cohen–Lenstra in Magma: Results



x-axis: prime $p \equiv 1 \pmod{4}$, $p \leq 10^6$. y-axis: cumulative proportion with $h_{\mathbb{Q}(\sqrt{p})} = 1$.

Cohen–Lenstra in Magma: Results



x-axis: prime $p \equiv 1 \pmod{4}$, $p \leq 10^6$. y-axis: cumulative proportion with $h_{\mathbb{Q}(\sqrt{p})} = 1$.

Conjectured limit: 0.7545. Observed at $p = 10^6$: 0.7840.

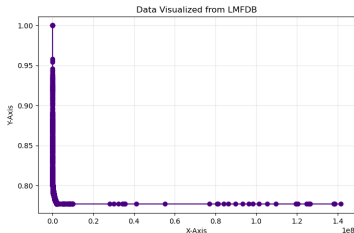
Idea. The LMFDB has already computed class numbers for all quadratic fields up to large discriminant. Just *query*, no need to recompute.

```
1 sage: from lmfdb import db          # also: lmfdb-lite from python
2 sage: query = {"degree": 2, "r2": 0, "num_ram": 1, # real, prime disc
3 ...          "disc_abs": {"$ne": 8}}          # "$ne" = postgres not-equal
4 sage: flds = list(db.nf_fields.search(query,["disc_abs", "class_number"]))
5 sage: cn = [f["class_number"] for f in flds]
6 sage: sum(c == 1 for c in cn) / len(cn) * 1.0
7 0.776977637552534
```

Cohen–Lenstra via LMFDB: $p \leq 10^8$

Idea. The LMFDB has already computed class numbers for all quadratic fields up to large discriminant. Just *query*, no need to recompute.

```
1 sage: from lmfdb import db          # also: lmfdb-lite from python
2 sage: query = {"degree": 2, "r2": 0, "num_ram": 1, # real, prime disc
3 ...           "disc_abs": {"$ne": 8}}           # "$ne" = postgres not-equal
4 sage: flds = list(db.nf_fields.search(query, ["disc_abs", "class_number"]))
5 sage: cn = [f["class_number"] for f in flds]
6 sage: sum(c == 1 for c in cn) / len(cn) * 1.0
7 0.776977637552534
```



Cumulative proportion with $h = 1$,
primes $p \equiv 1 \pmod{4}$, $p \leq 10^8$.

Conjectured: 0.7545
Magma (10^6): 0.7840
LMFDB (10^8): 0.7770

Even with $\sim 3 \cdot 10^6$ primes (LMFDB data up to 10^8), the proportion sits at 0.777 vs. conjectured 0.7545.

Even with $\sim 3 \cdot 10^6$ primes (LMFDB data up to 10^8), the proportion sits at 0.777 vs. conjectured 0.7545.

Slow convergence is the feature. The Cohen–Lenstra error term decays only logarithmically. This is precisely why testing the conjecture is hard – and why having LMFDB do the heavy lifting matters.

Even with $\sim 3 \cdot 10^6$ primes (LMFDB data up to 10^8), the proportion sits at 0.777 vs. conjectured 0.7545.

Slow convergence is the feature. The Cohen–Lenstra error term decays only logarithmically. This is precisely why testing the conjecture is hard – and why having LMFDB do the heavy lifting matters.

Open problem. Even the existence of infinitely many real quadratic fields with $h_K = 1$ (Gauss, 1801) is still unknown.

Hilbert Class Field and the LMFDB

The Hilbert Class Field

For a number field K , the **Hilbert class field** H_K is the maximal unramified abelian extension. By **class field theory**, $\text{Gal}(H_K/K) \simeq \text{Cl}(K)$.

```
1 > K<b> := QuadraticField(-23);
2 > H<a> := HilbertClassField(K);
3 > // Absolute degree 6, Galois group S_3
4 > Habs := AbsoluteField(H);
5 > Discriminant(Integers(Habs));
6 -12167 // = (-23)^3 = disc(K)^[H:K] -- characteristic of unramified
7 > IsIsomorphic(Habs, ext<K | x^3 - x - 1>);
8 true Mapping...
```

The Hilbert Class Field

For a number field K , the **Hilbert class field** H_K is the maximal unramified abelian extension. By **class field theory**, $\text{Gal}(H_K/K) \simeq \text{Cl}(K)$.

```
1 > K<b> := QuadraticField(-23);
2 > H<a> := HilbertClassField(K);
3 > // Absolute degree 6, Galois group S_3
4 > Habs := AbsoluteField(H);
5 > Discriminant(Integers(Habs));
6 -12167 // = (-23)^3 = disc(K)^[H:K] -- characteristic of unramified
7 > IsIsomorphic(Habs, ext<K | x^3 - x - 1>);
8 true Mapping...
```

The punchline. $H_{\mathbb{Q}(\sqrt{-23})} = \mathbb{Q}(\sqrt{-23})(\alpha)$ where $\alpha^3 - \alpha - 1 = 0$ – the running cubic from slide 3, with the same -23 ! The cubic was in our quadratic field's class group all along.

The Hilbert Class Field

For a number field K , the **Hilbert class field** H_K is the maximal unramified abelian extension. By **class field theory**, $\text{Gal}(H_K/K) \simeq \text{Cl}(K)$.

```
1 > K<b> := QuadraticField(-23);
2 > H<a> := HilbertClassField(K);
3 > // Absolute degree 6, Galois group S_3
4 > Habs := AbsoluteField(H);
5 > Discriminant(Integers(Habs));
6 -12167 // = (-23)^3 = disc(K)^[H:K] -- characteristic of unramified
7 > IsIsomorphic(Habs, ext<K | x^3 - x - 1>);
8 true Mapping...
```

The punchline. $H_{\mathbb{Q}(\sqrt{-23})} = \mathbb{Q}(\sqrt{-23})(\alpha)$ where $\alpha^3 - \alpha - 1 = 0$ – the running cubic from slide 3, with the same -23 ! The cubic was in our quadratic field's class group all along.

Bridge to Lecture 3. For imaginary quadratic K , there is also an analytic recipe: $H_K = K(j(\tau))$, where $\tau \in \mathcal{H}$ is a **CM** point (the elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ has $\text{End}(E_\tau) = \mathbb{Z}_K$). The minimal polynomial of $j(\tau)$ is the **Hilbert class polynomial** $H_{\Delta_K}(x)$ – explicit CFT via elliptic curves.

Sample queries you cannot do without LMFDB:

- Smallest $|\Delta_K|$ degree-5 field with Galois group A_5 (**5T4**): $|\Delta_K| = 18496 = 2^6 \cdot 17^2$.
- All number fields with class group $(\mathbb{Z}/2)^4$, sorted by discriminant.
- Number fields ordered by Galois group, discriminant, ramification type.
- Cross-references: number field $\rightarrow$ Galois representation $\rightarrow$ L -function $\rightarrow$ modular form.

Sample queries you cannot do without LMFDB:

- Smallest $|\Delta_K|$ degree-5 field with Galois group A_5 (**5T4**): $|\Delta_K| = 18496 = 2^6 \cdot 17^2$.
- All number fields with class group $(\mathbb{Z}/2)^4$, sorted by discriminant.
- Number fields ordered by Galois group, discriminant, ramification type.
- Cross-references: number field $\rightarrow$ Galois representation $\rightarrow$ L -function $\rightarrow$ modular form.

Lecture 1 callback: can you hear the shape of a number field? **Arithmetically equivalent** fields (Perlis 1977) are non-isomorphic number fields with the same Dedekind zeta function – the “spectrum” of a number field. Smallest example: degree 7, $|\Delta_K| = 13^2 \cdot 109^2$. *The drum-hearing failure of Lecture 1 has a number-field analogue.*

Sample queries you cannot do without LMFDB:

- Smallest $|\Delta_K|$ degree-5 field with Galois group A_5 (**5T4**): $|\Delta_K| = 18496 = 2^6 \cdot 17^2$.
- All number fields with class group $(\mathbb{Z}/2)^4$, sorted by discriminant.
- Number fields ordered by Galois group, discriminant, ramification type.
- Cross-references: number field $\rightarrow$ Galois representation $\rightarrow$ L -function $\rightarrow$ modular form.

Lecture 1 callback: can you hear the shape of a number field? Arithmetically equivalent fields (Perlis 1977) are non-isomorphic number fields with the same Dedekind zeta function – the “spectrum” of a number field. Smallest example: degree 7, $|\Delta_K| = 13^2 \cdot 109^2$. *The drum-hearing failure of Lecture 1 has a number-field analogue.*

Try this: https://www.lmfdb.org/NumberField/?galois_group=5T4

LMFDB + Magma: A Research Workflow

Each LMFDB number field page exports Magma-ready code.

```
1 > // Pasted from the LMFDB page for 5.5.18496.1
2 > R<x> := PolynomialRing(Rationals());
3 > K<a> := NumberField(x^5 - x^4 + 2*x^2 - 2*x + 2);
4 > // Now compute things LMFDB doesn't precompute:
5 > [<p, [<Norm(t[1]), t[2]> : t in Factorization(p*Integers(K))]>
6 >   : p in PrimesUpTo(20)];
7 > // Check Chebotarev: count totally split primes up to N
8 > N := 10^5;
9 > #[p : p in PrimesUpTo(N) | IsTotallySplit(p,Integers(K))] / (N/Log(N));
10 0.0176147759614044494827376346283
```

LMFDB + Magma: A Research Workflow

Each LMFDB number field page exports Magma-ready code.

```
1 > // Pasted from the LMFDB page for 5.5.18496.1
2 > R<x> := PolynomialRing(Rationals());
3 > K<a> := NumberField(x^5 - x^4 + 2*x^2 - 2*x + 2);
4 > // Now compute things LMFDB doesn't precompute:
5 > [<p, [<Norm(t[1]), t[2]> : t in Factorization(p*Integers(K))]>
6 >   : p in PrimesUpTo(20)];
7 > // Check Chebotarev: count totally split primes up to N
8 > N := 10^5;
9 > #[p : p in PrimesUpTo(N) | IsTotallySplit(p,Integers(K))] / (N/Log(N));
10 0.0176147759614044494827376346283
```

The pattern. Use LMFDB for browsing, reference data, and cross-references; use Magma for computations on top. This is how working number theorists actually use both tools together.

Summary

Key Takeaways

1. Number fields are governed by a small set of invariants: $\mathbb{Z}_K$, signature, discriminant, $\text{Cl}(K)$, $\mathbb{Z}_K^\times$.
2. **Kronecker-Weber**: abelian extensions of $\mathbb{Q}$ live inside cyclotomic fields; the conductor identifies the optimal one.
3. Magma computes Gauss-sum generators, conductors, class groups, and unit groups out of the box.
4. **Cohen-Lenstra heuristics** are open conjectures whose predictions can be tested computationally with a few lines of Magma.
5. The **Hilbert class field** packages the class group into a field extension — a bridge to CM theory and elliptic curves.
6. LMFDB + Magma together: browse, query, then compute on top.

Next lecture: elliptic curves, CM theory, and explicit class field theory of imaginary quadratic fields.